	Debre Birhan University		
1	College	**Computing**	**Department**: **Software Engineering**
2	Module Name	Computer Networking and Security	Module Code: 07
	Module Details		
	Course Name	**Principal of Distributed Systems**	
	Course Code	**SEng3074**	
3	Course Description	This course will focus on the principles, techniques, and practices relevant to the design and implementation of such systems. The course takes a systems-oriented view of distributed systems, concentrating on infrastructure software and providing hands-on experience of implementing distributed systems. Students will learn how to develop distributed applications using various programming languages and technologies, analyze and evaluate the performance of distributed systems, and understand the challenges and trade-offs involved in designing and deploying distributed systems.	
4	Name(s) of Academic Staff	Ms. Fikir Gizachew and Mr. Getachew Ganfur	
5	Semester and Year Offered	Semester: II	Year Offered: 3
6	Credit Hour	5 ECTS/ 3 Credit Hr.	
7	Prerequisite/Co-requisite	Operating Systems, Data Communications and Computer Networks	
8	Course Learning Outcome (CLO): At the end of the course, the student will be able to:		
	CLO1	Describe the design goals and key components of a distributed system and evaluate the tradeoffs of alternative architectural models.	
	CLO2	Explain the fundamental principles of distributed systems, including communication protocols, synchronization, and consistency models.	
	CLO3	Analyze and evaluate the performance of distributed systems, including scalability, fault tolerance, and load balancing	
	CLO4	Analyze the challenges and trade-offs involved in designing and deploying distributed systems, such as security, reliability, and data consistency.	
	CLO5	Design and implement distributed systems for various tasks including fault-tolerant design, load balancing, and data replication.	
	CLO6	Collaborate effectively in a team to design and implement distributed system projects.	

9 Mapping the Course Learning Outcomes to the Student Outcomes, Teaching Methods, and Assessment:

Course Learning Outcomes (CLO)	Student Outcomes (SOs)								Teaching Methods			Assessment					
	SO1	SO2	SO3	SO4	SO5	SO6	SO7	SO8	L	T	P	O	Mid-Exam	Project	Lab-Exam	Assignment	Final-Exam
CLO1	H	M	–	–	L	M	–	–	✓				✓				✓
CLO2	L	H	–	–	–	M	M	L	✓		✓		✓			✓	✓
CLO3	L	M	–	–	–	H	M	L	✓		✓			✓		✓	✓
CLO4	–	L	–	M	–	M	–	H	✓		✓			✓		✓	✓
CLO5	M	H	L	–	M	M	–	–	✓		✓		✓	✓			✓
CLO6	–	L	M	–	H	–	M	–			✓			✓			✓

High (H), Medium (M), Low (L), Blank (–): No meaningful contribution.

10	Transferable skills									
	Problem-solving, proficiency with distributed programming, .Net framework, and communication & teamwork									
11	Distribution of students' learning time (SLT)									
	Course content outline	**CLO**	Delivery time	Teaching learning Activities						Total
				Guided learning (F2F)				Guided learning (NF2F)	Independent Learning (NF2F)	
				L	T	P	O			
	Chapter 1: Introduction	**CLO1**	**Weeks 1**	**2**		**3**			**3**	**8**
	1.1. Introduction to distributed systems									
	1.2. Definition									
	1.3. Goals of a Distributed System									
	1.4. Types of Distributed Systems									
	2. Chapter 2 Architectures	**CLO1 CLO5**	**Week 2**	**2**		**3**			**3**	**8**
	2.1. Introduction									
	2.2. Architectural Styles									
	2.3. System Architectures									
	Lab contents: *Implementation of socket programming*									
	3. Chapter 3 Processes	**CLO1 CLO2 CLO5**	**Week 3-4**	**4**		**6**			**6**	**16**
	3.1. Introduction to Threads									
	3.2. Threads in Distributed									
	3.3. Systems									
	3.4. Clients									
	3.5. Servers									
	3.6. Code Migration									
	Lab contents: *Implementation of multi-threaded client/server processes, Write a Program to demonstrate process/code migration.*									
	4. Chapter 4: Communication	**CLO2 CLO4 CLO6**	**Week 5-6**	**4**		**6**			**6**	**16**
	4.1. Layer protocols									
	4.2. Types of Communication									
	4.3. Remote Procedure Call									

	4.4. Remote Object invocation									
	Lab contents: *Implementation of Client Server-based program using RMI, Implementation of Client Server-based program using RPC*									
	5. Chapter 5: Naming	**CLO2**	**Week 7-8**	**4**		**6**			**6**	**16**
	5.1. Naming Entities									
	5.2. Flat Naming									
	5.3. Structured Naming									
	5.4. Attribute-based naming									
	6. Chapter 6: Synchronization	**CLO2 CLO5 CLO6**	**Week 9-10**	**4**		**6**			**7**	**17**
	6.1. Overview									
	6.2. Clock Synchronization									
	6.3. Physical Clocks									
	6.4. Logical Clocks									
	Lab contents: Implementation of Clock Synchronization (logical/physical), Implementation of Election algorithm and Mutual Exclusion algorithms									
	7. Chapter 7: Consistency and Replication	**CLO2 CLO3**	**Week 11-12**	**4**		6			**6**	**16**
	7.1. Introduction to Consistency									
	7.2. Data Replication and Reasons for Replication									
	7.3. Replication as Scaling Technique									
	8. Chapter 8: Fault Tolerance	**CLO5 CLO6**	**Week 13-14**	**4**		**6**			**7**	**17**
	8.1. Basic Concepts									
	8.2. Failure Models									
	8.3. Process Resilience									
	8.4. Distributed Commit									
	8.5. Recovery									
	Lab contents: Write a distributed application using EJB, write a									

	program using CORBA to demonstrate object brokering, use .Net framework to deploy a distributed application									
	Mini Project: Integration of course concepts into a project; presentations & reflection		**Week 14**							
	Final Exam		**Week 15-16**						6	**6**
	Total			28		39	3		50	**120**

	Assessment				
	Continuous assessment		Percentage total (50%)	F2F	NF2F	SLT
	1	Mid-Semester Test (written)	20%	√		1
	2	Assignment	10%	√	√	3
	3	Course Project (team; code + demo + brief report)	20%	√	√	8
	Total					**12**
			Percentage total (50%)	F2F	NF2F	SLT
	Final exam		50%	√		3
	Grand Total					**135**
	L = Lecture, T = Tutorial, P = Practical, O = Others, F2F = Face-to-Face, NF2F = Non-Face-to-Face					

12	Special requirements and resources to deliver the course (e.g. software, computer lab, simulation room …etc.)	1	**Language Standard:** Java Programming Language
		2	**IDE / Editor:** NetBeans/IntelliJ IDEA
		3	**Lab Setup:** Computer lab with individual Windows workstations; projector for demonstrations.
13	Textbook	1	A. S. Tanenbaum and M. V. Steen, *Distributed Systems: Principles and Paradigms*, 2nd ed., Prentice Hall, 2007.
	Reference	1	G. F. Coulouris, J. Dollimore, and T. Kindberg, Distributed Systems: Concepts and Design, 5th ed., Addison-Wesley, 2005.
		2	S. Ghosh, Distributed Systems: An Algorithmic Approach, 2nd ed., Chapman & Hall/CRC, 2017.
		3	A. D. Kshemkalyani and M. Singhal, Distributed Computing: Principles, Algorithms, and Systems, Cambridge University Press, 2011.
		4	Hagit Attiya, Jennifer Welch. Distributed Computing: Fundamentals, Simulations, and Advanced Topics